Unit no. ° 10

VARIATIONS AND EXTREMA OF A FUNCTION

1-12 February 2027

I Increasing, decreasing and monotonic

Definitions

Over an interval I, f is increasing if for all $a, b \in I$, $a < b \Rightarrow f(a) \leq f(b)$; decreasing if $a < b \Rightarrow f(a) \geq f(b)$. The variations are strict when the inequalities between images are strict. A constant function has the same value everywhere; it is both increasing and decreasing in the broad sense. A monotonic function on I is increasing or decreasing on it.

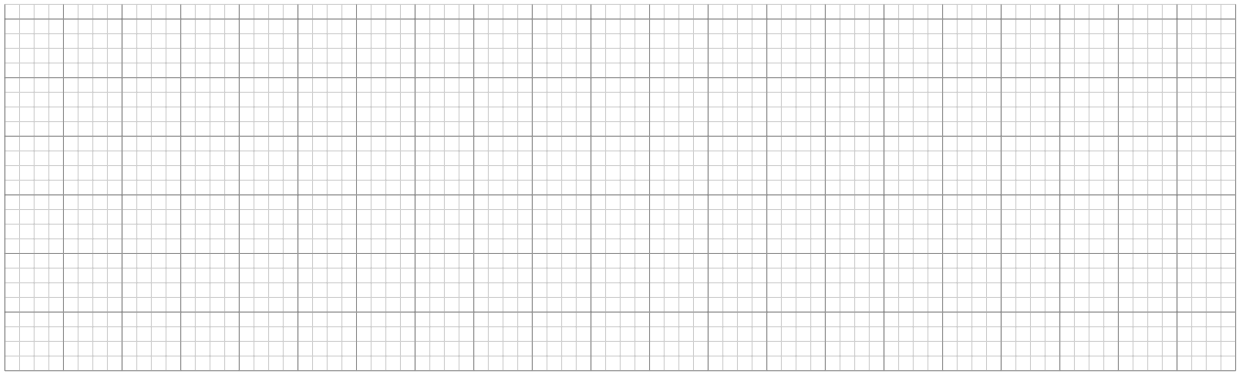
Table of variations

Give the domain, the intervals of monotonicity, the values at the endpoints when defined, then the arrows. A sign table shows whether $f(x)$ is positive or negative; a variation table indicates how its values evolve.

Example 10.1

On $[-3; 4]$, $f(x) = (x-1)^2 - 2$. Using the square, extending the curve studied in unit 6, draw its table of variations. Give two numbers $a < b$ such that $f(a) > f(b)$.

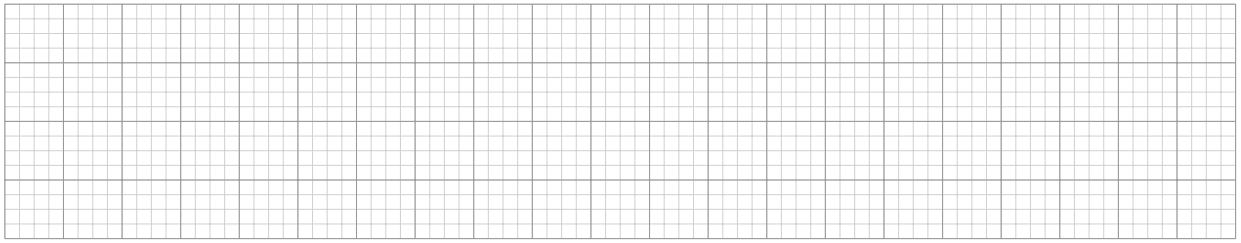
Solution:



Example 10.2

Can a function be negative and increasing? Give an example on an interval.

Solution:



II Maximum and minimum on an interval

Definitions

f has a maximum at $a \in I$ if, for all $x \in I$, $f(x) \leq f(a)$. The maximum value is $f(a)$.
f has a minimum at $b \in I$ if, for all $x \in I$, $f(x) \geq f(b)$. An extremum must be attained.
Distinguish its value from the x-value where it is attained.

Example 10.3

Revisit $f(x) = (x-1)^2 - 2$ on $[-3; 4]$. Give the minimum and maximum, then where they are attained. Redo on $[0; 4]$.

Solution:

Example 10.4

Does the fn $f(x) = x$ have a maximum on $[0; 1[$? A minimum?

Solution:

Read without inventing

A table of variations does not give all the intermediate images. It allows us to compare certain images; if the x-values lie on different branches, the comparison may be undetermined.

Example 10.5

Does the mere fact that a function decreases then increases allow us to compare two images on either side of the minimum?

Solution:

III Variations of affine functions

Property and rate of increase

For $f(x) = mx + p$ & $a \neq b$,

$$\frac{f(b) - f(a)}{b - a} = m.$$

If $m > 0$, f is strictly increasing; if $m < 0$, strictly decreasing; if $m = 0$, constant.

Write a proof.

Solution:

Example 10.6

Determine the variation of $f(x) = 3x + 2$, $g(x) = 7 - 6x$, $h(x) = -x$. Give the sign of $2x - 6$ and explain why the sign of this function does not remain constant even though it is increasing.

Solution:

Example 10.7

An affine function u satisfies $u(2) = 5$, $u(6) = 13$. Determine u, then its minimum and maximum on $[-1; 4]$.

Solution:

IV Variations of the square and absolute value functions

Properties

$x \mapsto x^2$ & $x \mapsto |x|$ are strictly decreasing on $]-\infty; 0]$ and strictly increasing on $[0; +\infty[$.
Their minimum on $\mathbb{R}$ is 0, reached only at 0.

Write a proof.

Solution:

Example 10.8

Compare without a calculator: $(-3,2)^2$ and $(-3,1)^2$; $1,2^2$ and $1,3^2$; $|-5|$ and $|-2|$.

Solution:

Example 10.9

Bound x^2 if $-3 \leq x \leq 2$. Why can we not simply square both bounds?

Solution:

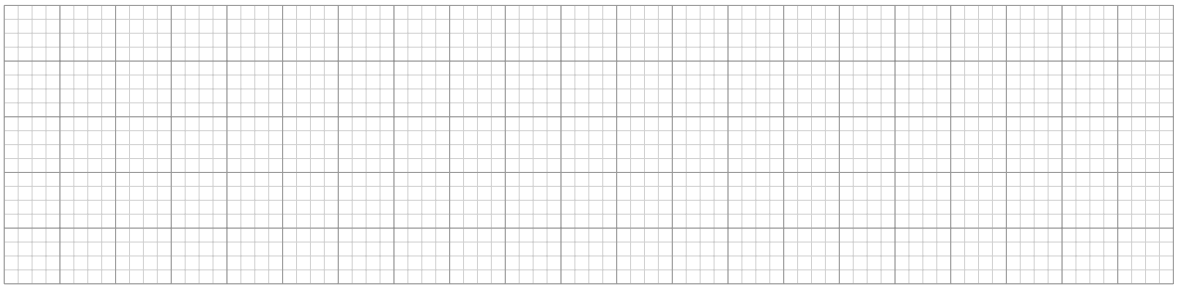
V Variations of the reciprocal function

Property

The reciprocal function is strictly decreasing on $]-\infty; 0[$ and on $]0; +\infty[$, separately. It is not decreasing on the whole of $\mathbb{R}^*$.

Write a proof.

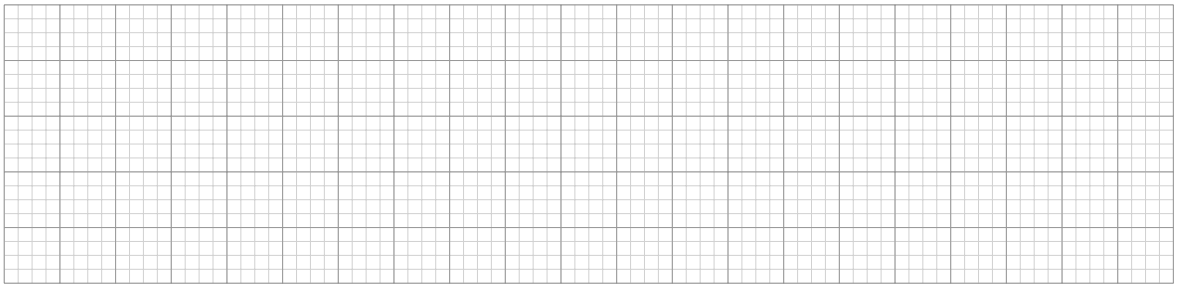
Solution:



Example 10.10

Compare $1/(-4)$ and $1/(-2)$, then $1/3$ and $1/5$. Bound $1/x$ if $2 \leq x \leq 5$, then if $-5 \leq x \leq -2$.

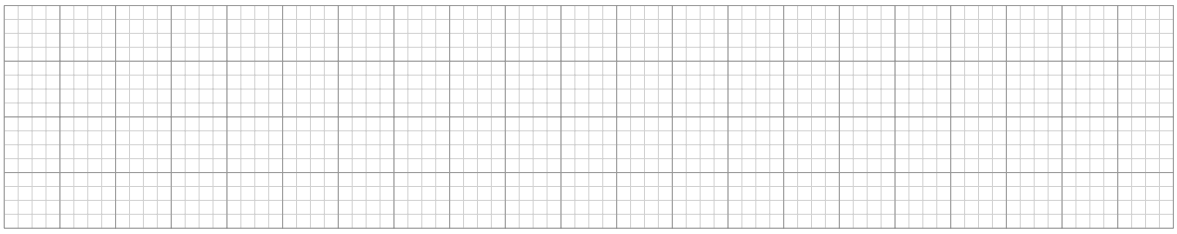
Solution:



Example 10.11

Why is the interval $[-1; 1]$ not suitable for directly applying a variation rule for the reciprocal function?

Solution:



VI Relative position of a number and its square

Property for positive numbers

For $0 \leq x \leq 1$, $x^2 \leq x$; for $x \geq 1$, $x^2 \geq x$. Equality holds at 0 and 1.

Write a proof.

Solution:

Square root and cube

The square root function is strictly increasing on $[0; +\infty[$. The cube function is strictly increasing on $\mathbb{R}$.

Example 10.12

Explain why the square root is increasing from the fact that squaring is increasing on the positive numbers. For the cube, use $b^3 - a^3 = (b-a)(a^2+ab+b^2)$.

Solution:

Example 10.13

Compare $\sqrt{5}$ and $\sqrt{6}$, then $(-2)^3$ and $(-1)^3$.

Solution:

VII Optimise: conjecture and prove

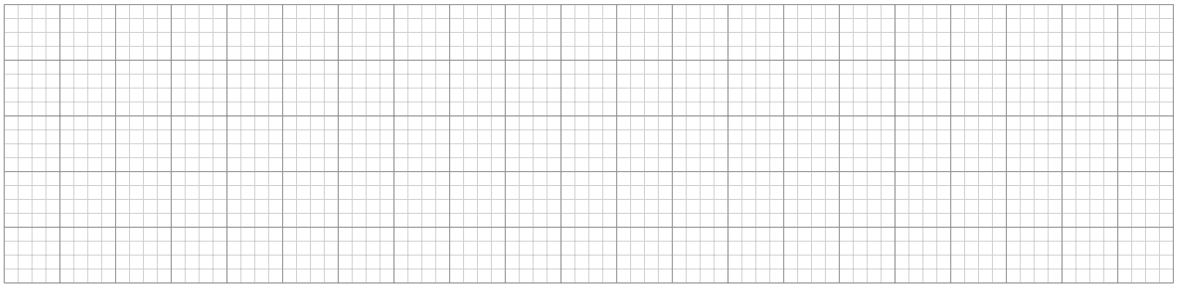
Example 10.14

A rectangle with perimeter 20 m has area $A(x) = x(10 - x)$ with $x \in]0; 10[$.

1. Conjecture the maximum area with a table of values.

2. Show $A(x) = 25 - (x - 5)^2$. Deduce the exact maximum and the corresponding dimensions.

Solution:



```
def aire(x):
    return x*(10-x)

def maximum_grille(N):
    # N integer >= 2 ; points strictly between 0 and 10
    meilleur_x = 10/N
    meilleur_y = aire(meilleur_x)
    for k in range(2, N):
        x = 10*k/N
        if aire(x) > meilleur_y:
            meilleur_x, meilleur_y = x, aire(x)
    return meilleur_x, meilleur_y
```

Example 10.15

What does the program return for $N = 10$? Does a numerical scan always prove the maximum of a function over an interval?

Solution:

